



Data Warehousing and
Business Intelligence
(IT3021)

Assignment 2

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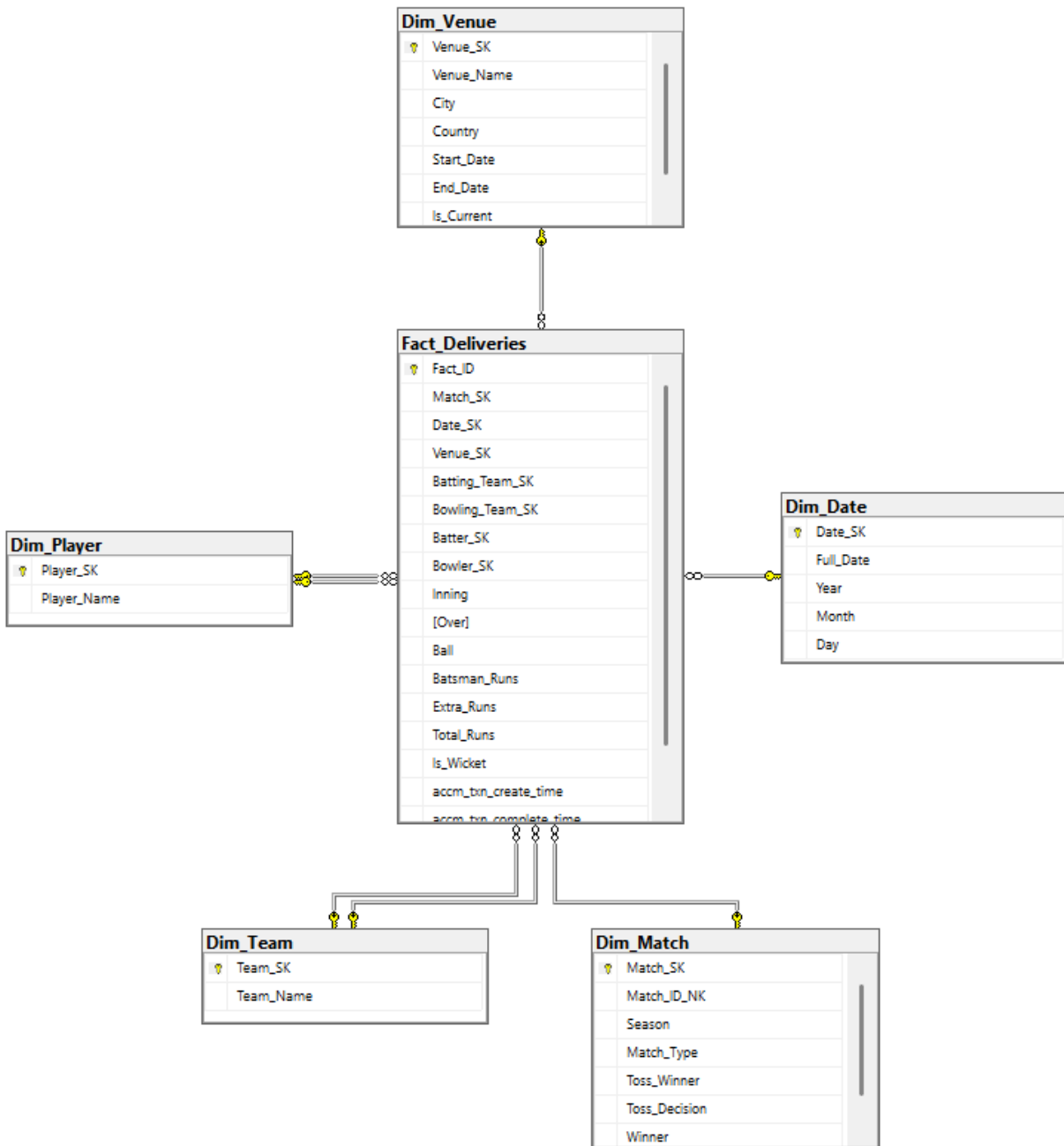
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Step 1: Data source for the Assignment 2

Overview of the Data Source

For this reporting phase, the data warehouse designed, implemented, and loaded in Assignment 1 serves as the primary data source. The data warehouse is built upon a **Star Schema** architecture, where a central fact table is directly surrounded by its associated dimension tables. This multidimensional model is highly optimized for analytical querying, spatial analysis, and time-series reporting. To maintain historical integrity and isolate the analytical environment from source system alterations, all dimensional tables utilize auto-incrementing Surrogate Keys (Primary Keys), which map to corresponding Foreign Keys in the fact table.

Entity-Relationship (ER) Diagram



Schema Description and Granularity

The data model consists of one central fact table and five surrounding dimension tables, structured as follows:

- **Fact Table (Fact_Deliveries):** This is the core transactional table of the warehouse, storing granular metrics such as runs and wickets.
 - **Grain Definition:** The granularity of this fact table is explicitly defined at the lowest possible level: **one row per single ball bowled in an innings.**
- **Dimension Tables:**
 1. **Dim_Venue (SCD Type 2):** Connects to the fact table to enable spatial data analysis (e.g., City, Country). Crucially, this is implemented as a **Slowly Changing Dimension (Type 2)**. By utilizing Start_Date, End_Date, and Is_Current attributes, it accurately tracks historical updates—such as stadium name changes due to sponsorships—ensuring past match records reflect the correct venue name at that specific time.
 2. **Dim_Team (Role-Playing Dimension):** Contains team attributes. It functions as a **Role-Playing Dimension** by connecting to Fact_Deliveries twice (via Batting_Team_SK and Bowling_Team_SK). This design allows for the independent and simultaneous analysis of both the batting and bowling franchises within the exact same delivery.
 3. **Dim_Player:** Connected to analyze the individual performances of batsmen and bowlers.
 4. **Dim_Date:** Enables robust time-series analysis, allowing performance evaluations across different years, months, and seasons.
 5. **Dim_Match:** Provides context at the match level, storing match summaries and broader season details.

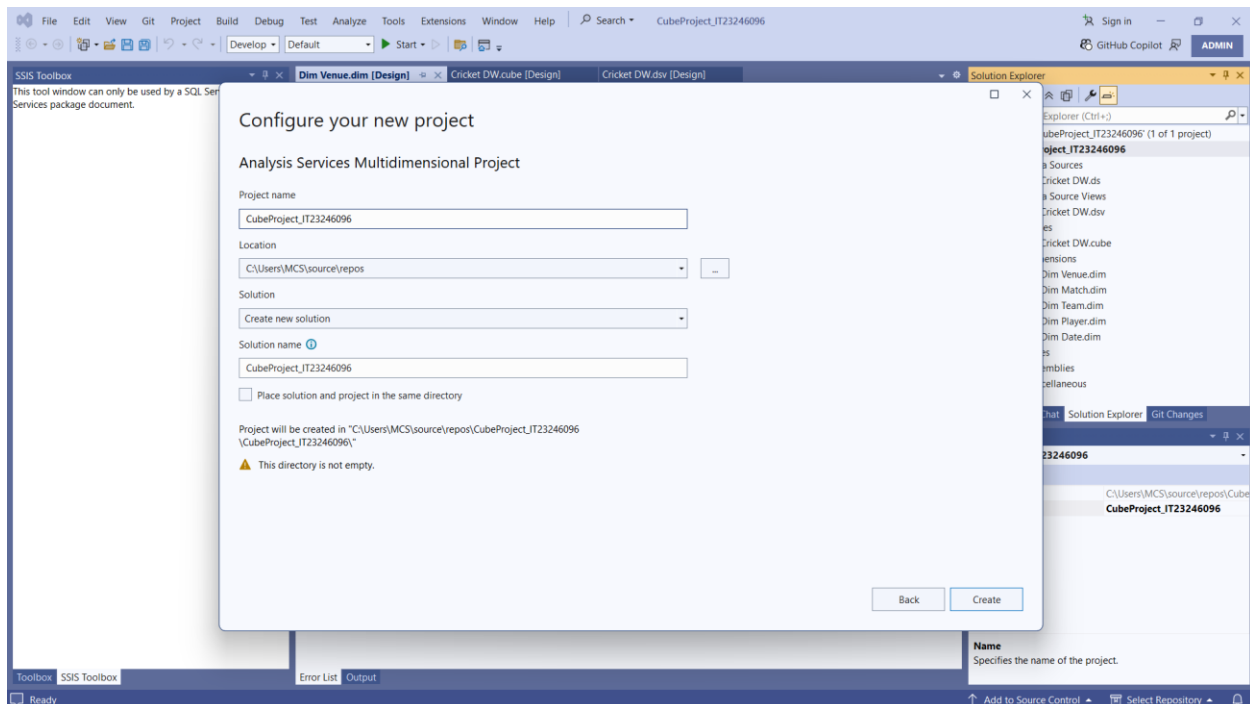
Step 2: SSAS Cube Implementation

Overview

The objective of this phase was to design and implement a SQL Server Analysis Services (SSAS) Multidimensional Cube using the previously constructed Data Warehouse (Cricket_DW) as the underlying data source. The cube facilitates Online Analytical Processing (OLAP), enabling high-performance, multidimensional analysis of cricket match statistics. Key requirements fulfilled in this step include defining relevant measures from the fact table, linking all associated dimension tables, and establishing at least one user-defined hierarchy to support drill-down capabilities.

Project Initialization

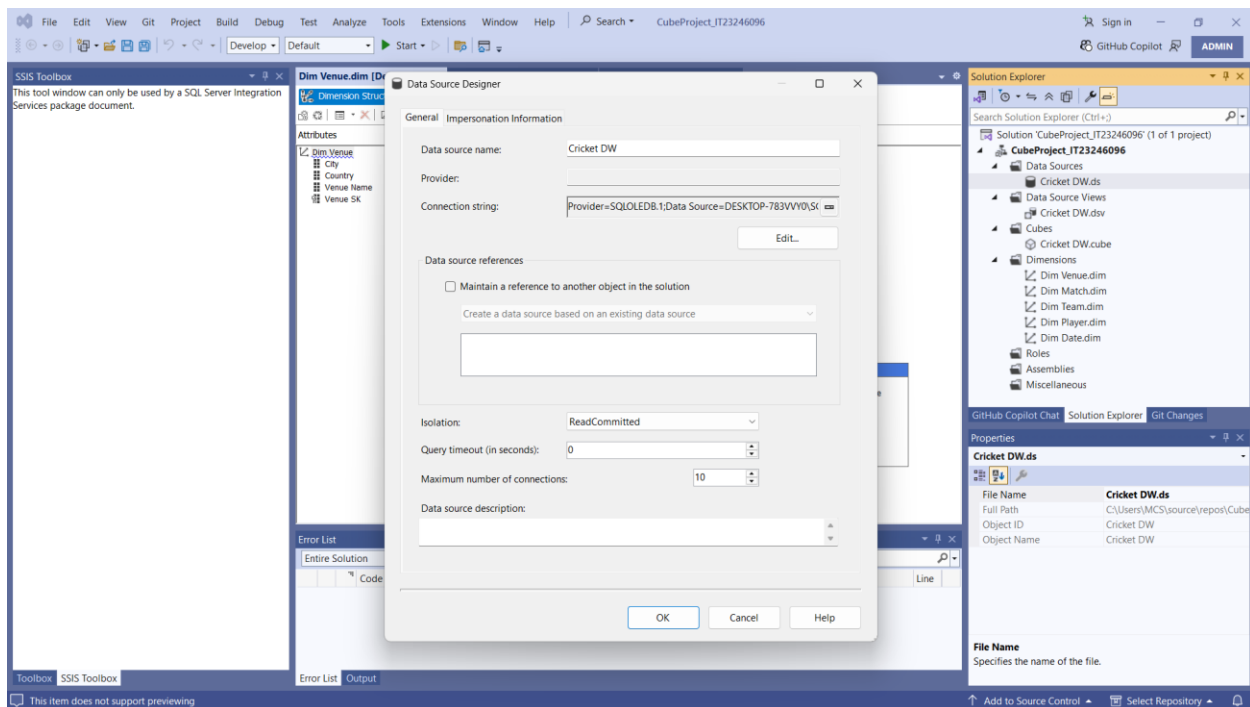
The development process began by launching Visual Studio (equipped with SQL Server Data Tools - SSDT). A new project was instantiated using the **Analysis Services Multidimensional and Data Mining Project** template. Following the assignment naming conventions, the project was designated as CubeProject_IT23246096.



Configuring the Data Source

To bridge the SSAS project with the underlying relational data warehouse, a new Data Source was configured.

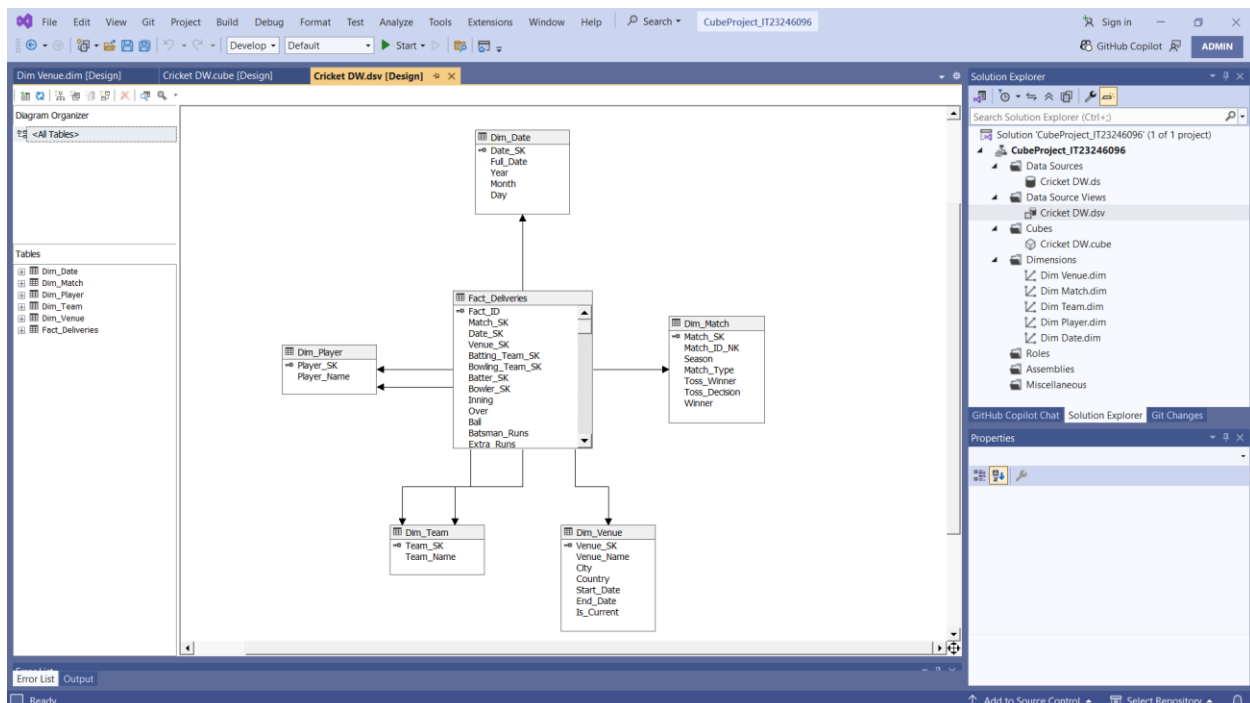
1. Within the Solution Explorer, a new Data Source was created pointing to the local SQL Server instance (.\SQL2022).
2. The specific database selected as the source was Cricket_DW.
3. For security and access control during processing, the Impersonation Information was configured to utilize a specific service account (or 'Use the service account' depending on your specific successful configuration), ensuring the OLAP engine possessed the necessary read permissions (db_datareader) to access the relational tables.



Defining the Data Source View (DSV)

A Data Source View (DSV) acts as a logical abstraction layer over the physical database. It was created to select only the tables relevant to the multidimensional model.

1. The previously created Data Source was selected as the foundation.
2. The central fact table, Fact_Deliveries, along with all associated dimension tables (Dim_Match, Dim_Venue, Dim_Player, Dim_Team, and Dim_Date) were moved from the 'Available objects' pane to the 'Included objects' pane.
3. Upon completion, the DSV automatically inferred the foreign key relationships established during the ETL phase, rendering a visual Star Schema representation.



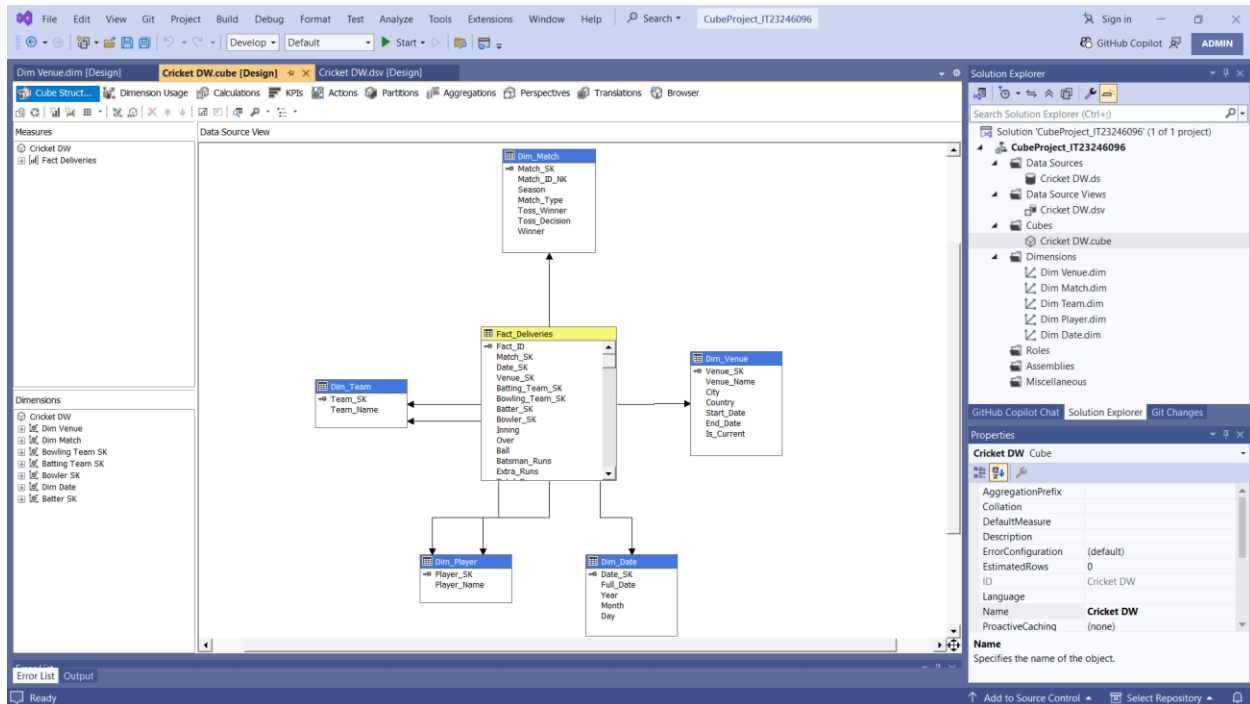
Cube Design and Configuration

With the DSV established, the actual multidimensional cube was generated.

1. Utilizing the Cube Wizard, the option to 'Use existing tables' was selected.
2. **Identifying Measures:** The Fact_Deliveries table was exclusively selected to define the Measure Group. The wizard automatically identified numeric columns as potential measures. The final measures included key performance indicators such

as Batsman Runs, Extra Runs, and Total Runs (aggregated using the SUM function) to support quantitative analysis.

3. **Identifying Dimensions:** The wizard correctly identified Dim_Match, Dim_Venue, Dim_Player, Dim_Team, and Dim_Date as the dimensions required to slice and dice the factual data.

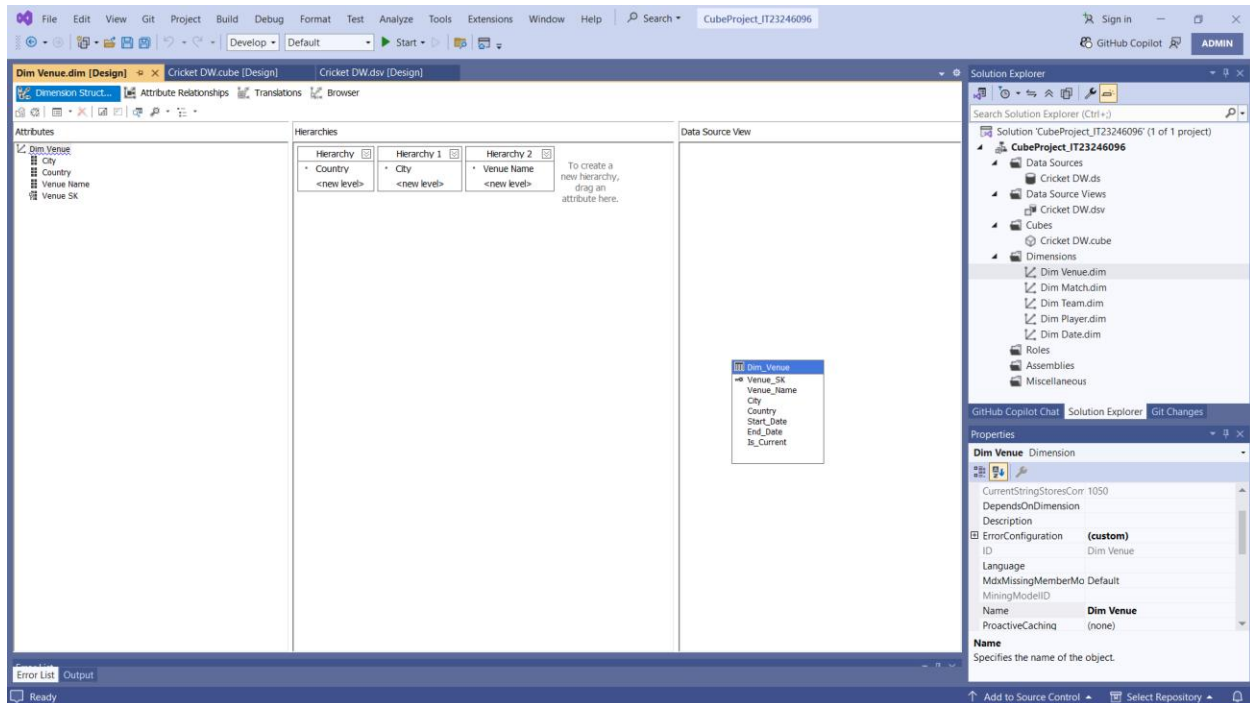


Establishing Dimension Hierarchies

To fulfill the specific assignment requirement and to enhance the analytical capabilities of the cube, a geographic hierarchy was implemented within the Venue dimension.

Hierarchies allow end-users to seamlessly drill down from aggregated summaries to granular details.

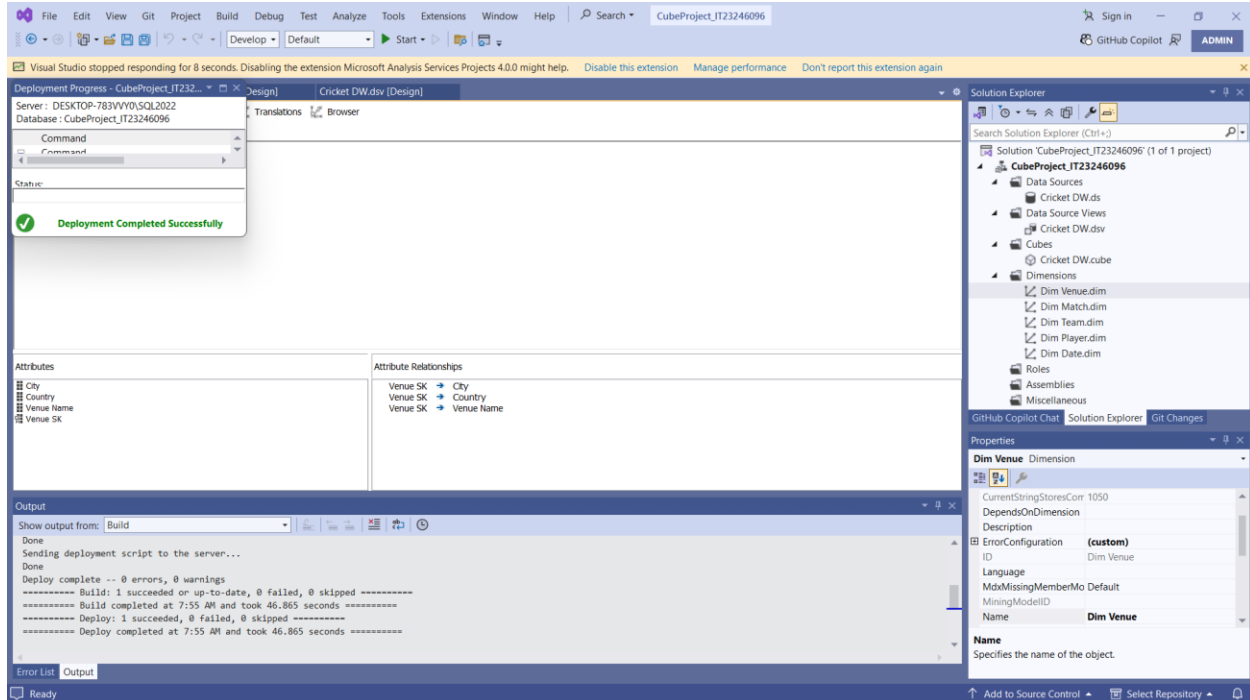
1. The Dim_Venue.dim file was opened in the Dimension Designer.
2. The attributes Country, City, and Venue Name were dragged from the Data Source View into the Attributes pane.
3. In the Hierarchies pane, a new user-defined hierarchy was constructed.
4. The levels were organized top-down: Country -> City -> Venue Name, representing the natural geographic drill-down path.



Deployment and Processing

The final step involved deploying the structural metadata to the Analysis Services server and processing the cube to populate it with data from the relational warehouse.

1. The Target Server property for the project deployment was verified and set to the correct named instance (.\\SQL2022).
2. The project was successfully deployed to the server without errors.
3. Subsequently, a **Process Full** operation was executed on the cube. This operation extracted the data from Cricket_DW, built the aggregations, and populated the multidimensional structures.
4. The processing completed successfully, confirming that all permissions (including the previously resolved NT SERVICE\\MSOLAP\$\\SQL2022 login issue) and relationships were correctly configured.



Step 3: Demonstration of OLAP Operations

Overview

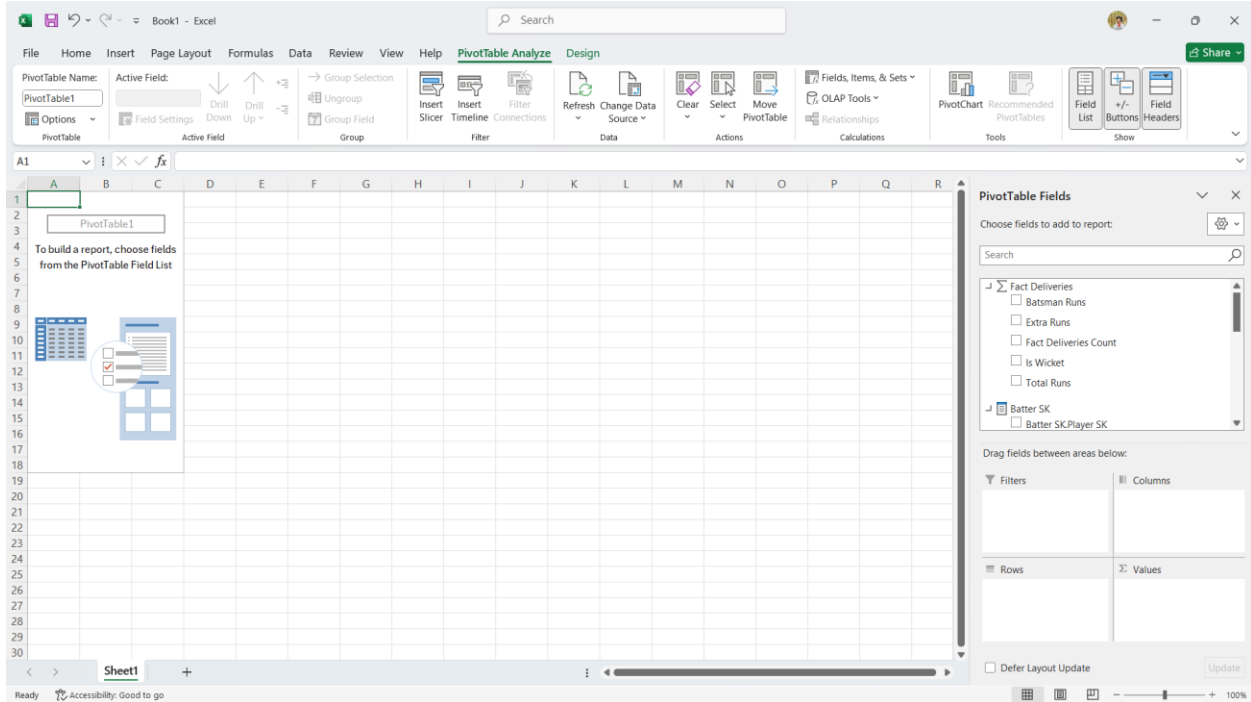
The final phase of this project involves connecting a front-end analytical tool to the deployed SSAS multidimensional cube to perform Online Analytical Processing (OLAP). Microsoft Excel was utilized to connect to the cube and demonstrate five fundamental OLAP operations: Roll-up, Drill-down, Slice, Dice, and Pivot.

Connecting Excel to the SSAS Cube

To begin the analysis, a live connection was established between MS Excel and the SQL Server Analysis Services instance.

1. Opened a blank Excel workbook and navigated to the **Data** tab.
2. Selected **Get Data > From Database > From Analysis Services**.
3. Entered the local server name (.\SQL2022 / DESKTOP-783VVY0\SQL2022) and used Windows Authentication.

4. Selected the CubeProject_IT23246096 database and the corresponding multidimensional cube.
5. Imported the data as a **PivotTable Report** to enable dynamic multidimensional analysis.



Executing OLAP Operations

Through the Excel PivotTable interface, the following standard OLAP operations were successfully executed using the Cricket_DW multidimensional dataset:

Roll-up

Definition: Roll-up involves summarizing or aggregating data along a dimension hierarchy (moving up from more detailed data to less detailed data).

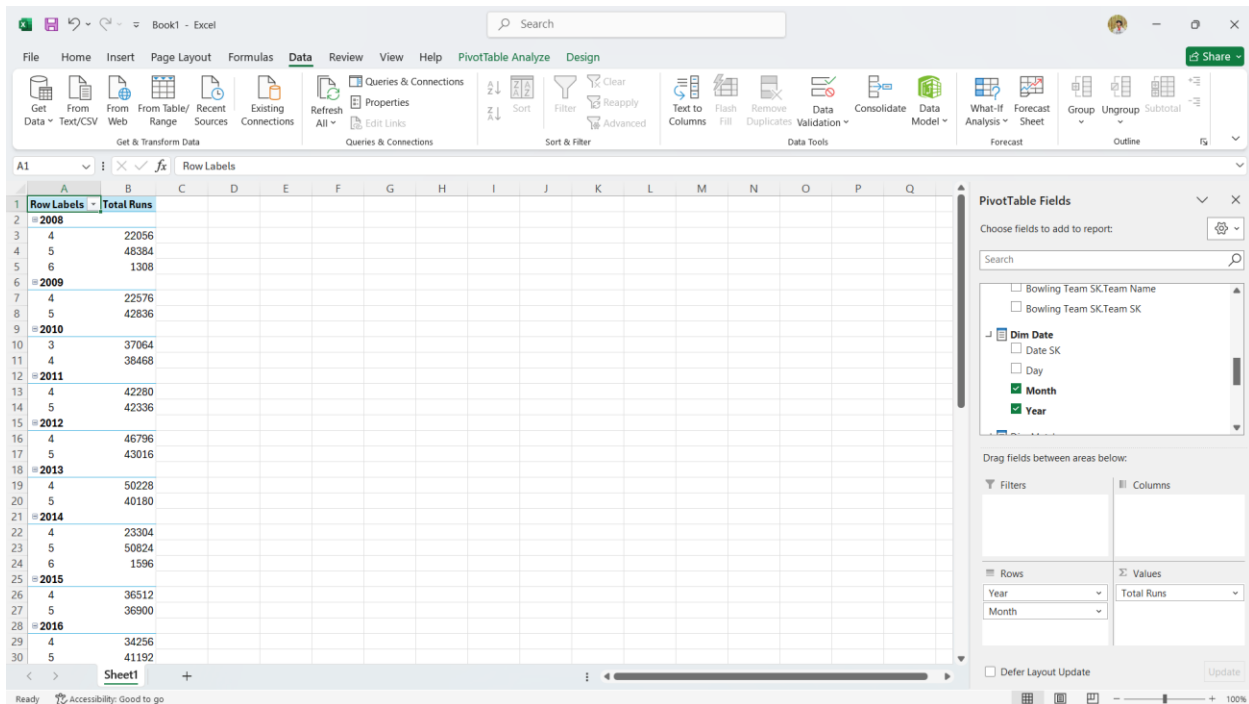
Implementation: I utilized the Dim Date dimension to view the summarized Total Runs (from Fact Deliveries) grouped by Year. This operation aggregates all the runs scored in individual matches and months into a high-level yearly summary.

Row Labels	Total Runs
2008	71748
2009	65412
2010	75532
2011	84616
2012	89812
2013	90408
2014	75724
2015	73412
2016	75448
2017	75144
2018	79604
2019	77736
2020	77664
2021	74548
2022	97580
2023	102752
2024	103884
Grand Total	1391024

Drill-Down

Definition: Drill-down is the reverse of roll-up. It involves navigating from a highly aggregated level of data down to a more granular level to reveal finer details.

Implementation: Using the time hierarchy established in the Dim Date dimension, I expanded a specific Year node to reveal the Month level data. This breaks down the yearly summarized runs into monthly statistics, providing deeper insights into the scoring patterns within that year.



Row Labels	Total Runs
2008	
4	22056
5	48384
6	1308
2009	
4	22576
5	42836
2010	
3	37064
4	38468
2011	
4	42280
5	42336
2012	
4	46796
5	43016
2013	
4	50228
5	40180
2014	
4	23304
5	50824
6	1596
2015	
4	36512
5	36900
2016	
4	34256
5	41192

Slice

Definition: The Slice operation performs a selection on one dimension of the given cube, resulting in a sub-cube. It essentially acts as a filter on a single attribute.

Implementation: I applied a filter on the Batting Team SK dimension. By selecting a specific team (e.g., Chennai Super Kings) from the filter dropdown, the entire PivotTable was "sliced" to display the total runs solely for that chosen team, isolating their performance from the rest of the dataset.

The screenshot shows an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable is filtered by 'Batting Team SK' to show data for 'Deccan Chargers'. The task pane shows the 'Batting Team SK' dimension selected in the Filters area.

Batting Team SK	Deccan Chargers
2008	
4	2484
5	6432
2009	
4	3092
5	6540
2010	
3	3840
4	5656
2011	
4	4252
5	4308
2012	
4	4052
5	5196
Grand Total	45852

PivotTable Fields

Choose fields to add to report:

- Batting Team SK (Selected)
- Bowler SK
- Bowler SK.Player Name
- Bowler SK.Player SK
- Bowling Team SK
- Bowling Team SK.Team Name

Drag fields between areas below:

Filters	Columns
Batting Team SK	

Rows	Values
Year	Total Runs
Month	

Defer Layout Update: ☐ Update

Dice

Definition: The Dice operation defines a sub-cube by performing a selection on two or more dimensions simultaneously.

Implementation: Building upon the Slice operation, I introduced a second dimension filter. I filtered the data by a specific Batting Team AND a specific Venue Name (from Dim Venue). This dual-condition filtering creates a highly specific sub-cube, showing the performance of a particular team at a particular stadium.

The screenshot shows an Excel spreadsheet with a PivotTable and the PivotTable Fields task pane. The PivotTable is titled 'Batting Team SK.Team Name' and is located in the range A1:B3. The PivotTable has two columns: 'Batting Team SK.Team Name' and 'Total Runs'. The data is filtered by 'Batting Team SK.Team Name' (Deccan Chargers) and 'Venue Name' (Barabati Stadium). The PivotTable shows the following data:

Batting Team SK.Team Name	Total Runs
2010	1364
2012	504
5	744
Grand Total	2612

The PivotTable Fields task pane is open on the right side of the screen. It shows the following fields:

- Filters: Batting Team SK.Team, Venue Name
- Columns: (Empty)
- Rows: Year, Month
- Values: Total Runs

The task pane also includes a search bar and a 'Defer Layout Update' checkbox.

Pivot

Definition: Pivot (or rotate) changes the dimensional orientation of a report or page display. It rotates the data axes to provide an alternative presentation of the data.

Implementation: I restructured the PivotTable layout by swapping the dimensions on the axes. The Year attribute was moved from the 'Rows' area to the 'Columns' area, and the Batting Team Name was moved to the 'Rows' area. This rotation instantly transformed the visualization, allowing for a comparative cross-tabular analysis of team performances across different years.

The screenshot shows an Excel spreadsheet with a PivotTable titled 'Total Runs'. The PivotTable is structured with 'Venue Name' as the filter, 'Year' as the column labels, and 'Team Name' as the row labels. The values are 'Total Runs'. The data is summarized for three years: 2010, 2012, and 2014, with a Grand Total. The PivotTable Fields task pane on the right shows the field list with 'Venue Name' selected for Filters, 'Year' for Columns, 'Month' for Rows, and 'Total Runs' for Values. The 'Defer Layout Update' checkbox is checked.

	2010	2012	2014	Grand Total
Deccan Chargers	1364			1364
Delhi Daredevils	644			644
Kings XI Punjab	656			656
Deccan Chargers		504		504
Kolkata Knight Riders		508		508
Deccan Chargers		744		744
Kings XI Punjab		1520		1520
Kolkata Knight Riders		1168		1168
Mumbai Indians		564		564
Pune Warriors		692		692
Unknown		748		748
Grand Total	2664	2448	4000	9112

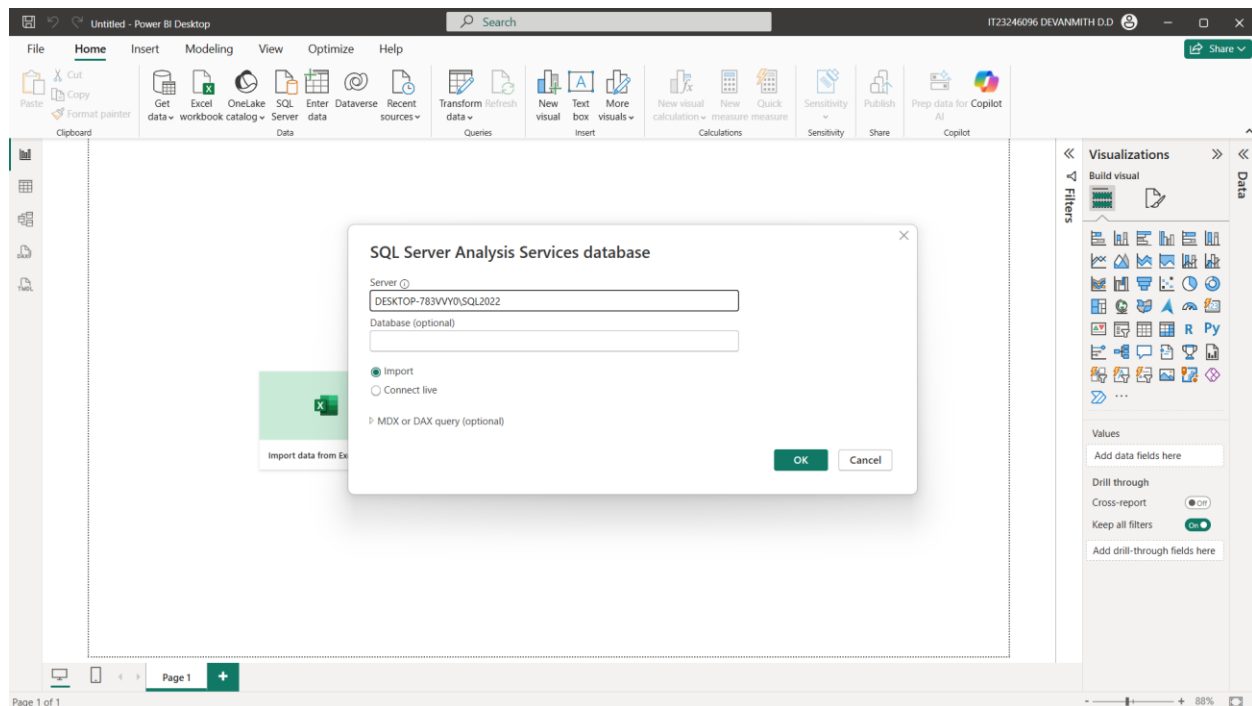
Step 4: PowerBI Reports

Data Preparation & Modeling

For the reporting phase, Microsoft Power BI was used to visualize the analytical data. The data was imported from the deployed SQL Server Analysis Services (SSAS) Multidimensional Cube (CubeProject_IT23246096) into Power BI's in-memory storage.

This approach allows the report to leverage the pre-defined star schema, relationships, and dimensions from the SSAS cube while ensuring high-performance interactivity and data

exploration within the Power BI environment. Additionally, implicit DAX aggregations were utilized directly from the imported cube measures to construct the visual summaries efficiently.



Report Design and Configurations

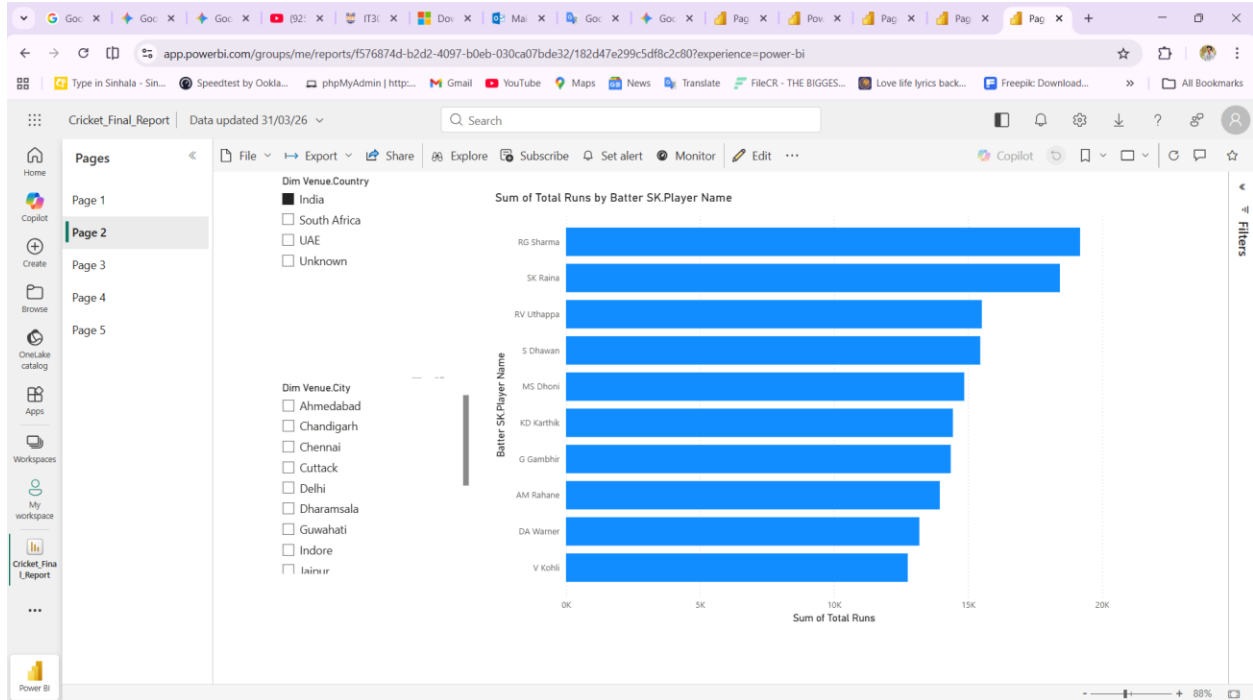
Report 1: Detailed Matrix Visual

- **Objective:** To display tabular data with clear row and column groupings.
- **Configuration:** A Matrix visual was utilized. The Batting Team Name from the Dim Team dimension was assigned to the Rows, while the Season (Year) from the Dim Date dimension was assigned to the Columns. The Total Runs measure from the Fact Deliveries table was used for Values, creating a comprehensive cross-tabular view of team performances over the years.

Batting Team SK Team Name	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	Total
Deccan Chargers	8916	9632	9496	8560	9248													45852
Delhi Capitals												10116	11120	9980	9364	8728	10292	59600
Delhi Daredevils	8472	8524	8620	8348	10580	8980	8324	8672	8416	8876	9372							97184
Gujarat Lions									9800	9648								19448
Gujarat Titans													10652	12216	8160			31028
Kings XI Punjab	9856	7712	9152	8896	9560	9712	11948	8072	8536	8828	8840	9716	9428					120256
Kochi Tuskers Kerala				7604														7604
Kolkata Knight Riders	7768	7088	8576	8092	10016	9160	9996	8176	9052	9936	11036	9892	8888	10236	8892	9852	10668	157324
Lucknow Super Giants														10192	9916	9932		30040
Mumbai Indians	8320	7588	10952	9152	9848	11900	9412	10936	8776	11200	9520	10680	11032	8468	8868	11780	10272	168704
Pune Warriors				7100	9284	9048												25432
Punjab Kings														8600	9372	10224	9948	38144
Rajasthan Royals	10404	6824	8716	6748	10064	10820	8664	8572			9428	8776	9152	8784	11772	9676	10588	138988
Rising Pune Supergiant										9880								9880
Rising Pune Supergiants										8252								8252
Royal Challengers Bangalore	7932	9120	9604	9876	9888	10404	8372	9244	11924	7380	9288	8832	9156	9212	10528	10008		150768
Royal Challengers Bengaluru																	11720	11720
Sunrisers Hyderabad						9272	8408	8840	10692	9396	10884	9832	10124	8336	8788	9504	12208	116284
Unknown	10080	8924	10416	10240	11324	11112	10600	10900		11236	9892	8764	10932	9152	10848	10096		154516
Total	71748	65412	75532	84616	89812	90408	75724	73412	75448	75144	79604	77736	77664	74548	97580	102752	103884	1391024

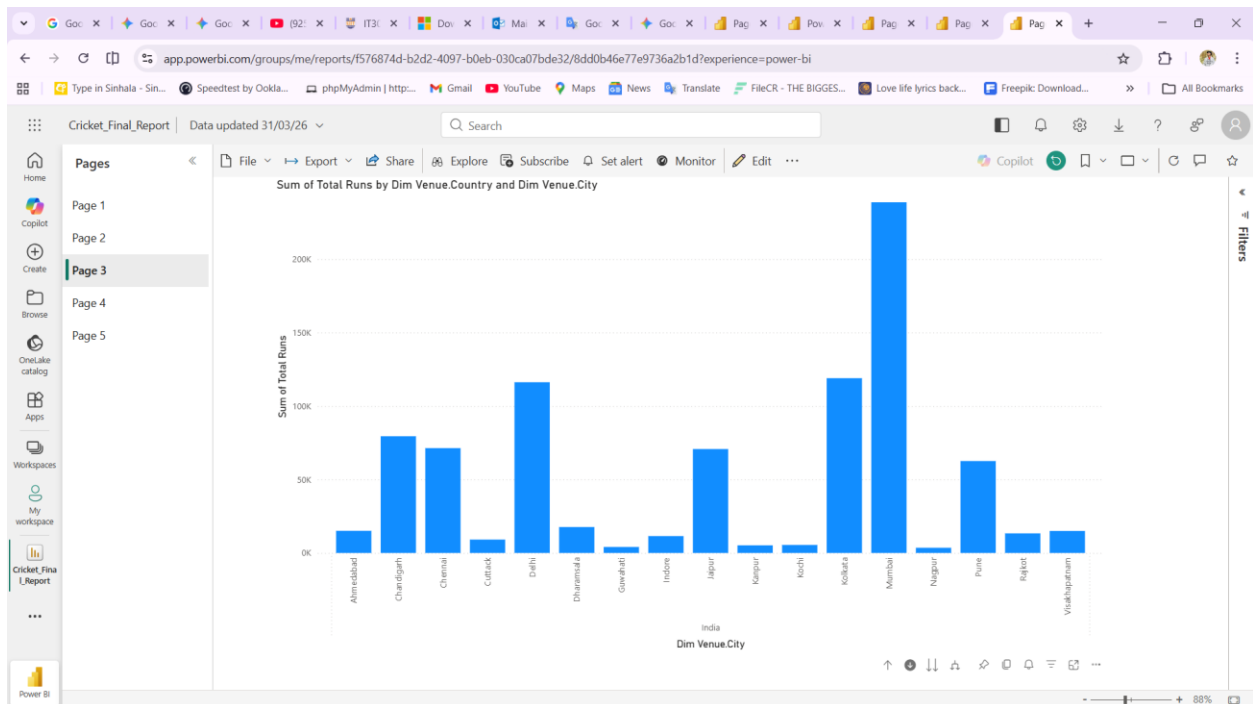
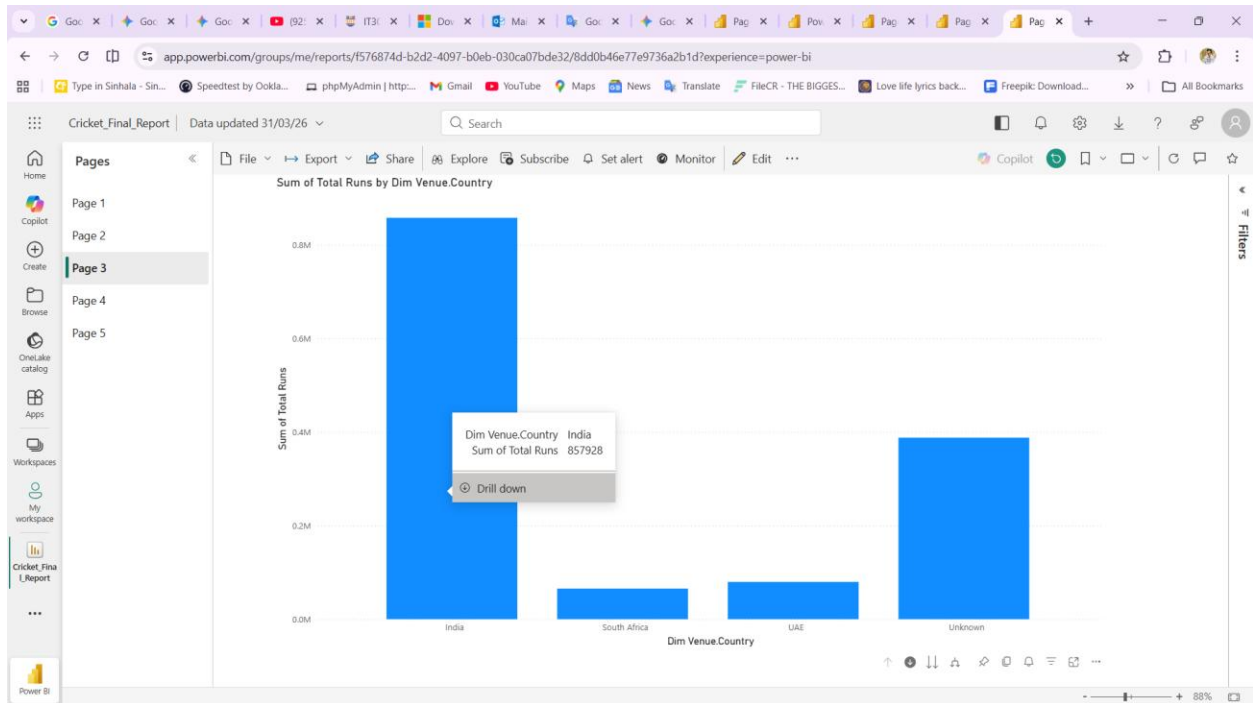
Report 2: Cascading Slicers and Dynamic Insights

- Objective:** To implement dynamic filtering using cascading parameters alongside analytical charts.
- Configuration:** Two Slicer visuals were introduced. The first slicer was bound to Country and the second to City (both from Dim Venue). Due to the hierarchical nature of the SSAS cube, selecting a Country dynamically filtered the available options in the City slicer (Cascading effect). A Clustered Bar Chart was added to display the Top 10 Batsmen by Total Runs, which updates dynamically based on the slicer selections.



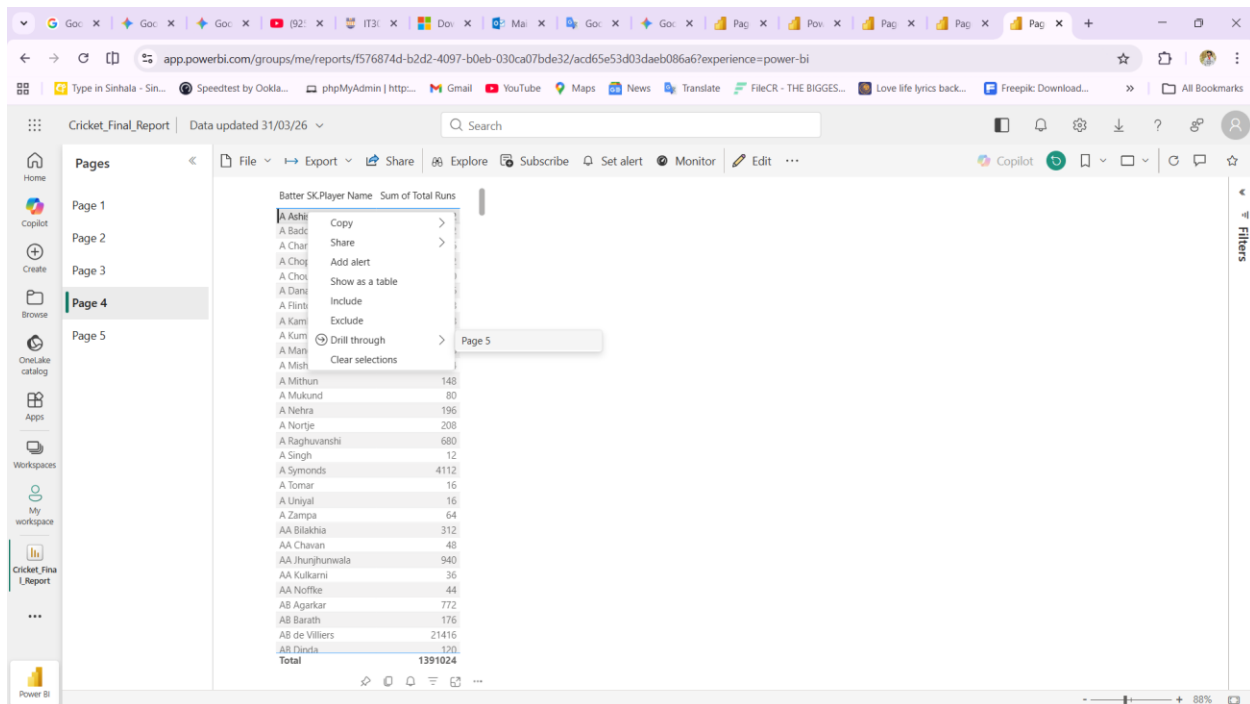
Report 3: Hierarchical Drill-Down Report

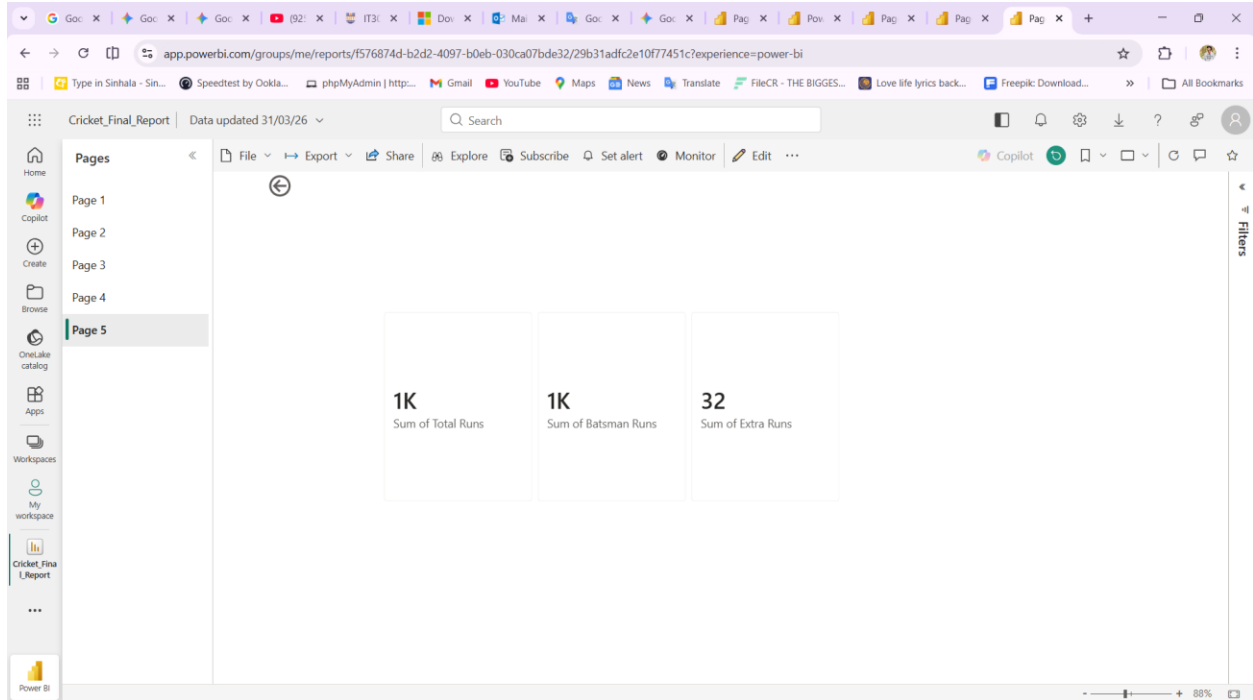
- **Objective:** To allow users to navigate data from a macro to a micro level seamlessly.
- **Configuration:** A Clustered Column chart was created utilizing the geographical hierarchy: Country -> City -> Venue Name on the X-axis, with Total Runs on the Y-axis. By enabling the "Drill-down" feature, clicking on a specific Country expands the visual to display the runs distributed across its respective Cities, and further down to specific Venues.



Report 4: Drill-Through for Detailed Insights

- **Objective:** To navigate from a high-level summary page to a highly detailed specific page based on user selection.
- **Configuration:** This required a two-page setup. A 'Details Page' was created where Batter Name was placed in the "Drill-through fields" section. Key Performance Indicator (KPI) cards were added to this page to show specific metrics (e.g., Batsman Runs, Total Runs). On the 'Main Page', a table listing all Batsmen was created. Users can right-click any specific player name and select "Drill through" to instantly navigate to the Details Page, which is automatically filtered to show only that selected player's statistics.





Publishing to Power BI Service

All developed reports and dashboards were successfully published to the **Power BI Service (online)** from Power BI Desktop. This deployment ensures that the analytical insights are hosted in the cloud environment, fulfilling the requirement for online accessibility and demonstration.